Linear Algebra I

FMA_1F001_EP

Lecture Notes

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These are personal lecture notes and may contain errors. For the official course material consult the lecturer.

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1 Lecture I: Groups. Vector Space

Definition 1: Group

A *group* is a (non-empty) set G, together with an operation $*$: $G \times G \longrightarrow G$, $(g,h) \longmapsto g*h$, Satisfying:

- Neutral/Identity element $\exists e \in G$ s.t. $\forall g \in G$: $e * g = g * e = g$,
- Associativety $\forall g_1, g_2, g_3 \in G$: $(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$,
- Inverses/Opposites $\forall g \in G \exists h \in G$: $g * h = h * g = e$.

Furthermore, we say G is *Abelian* or *Commutative* if $\forall g_1, g_2 \in G$: $g_1 * g_2 = g_2 * g_1$.

Example 1

- $G = \{0\}$, with $0 + 0 = 0$ (zero group),
- $\mathbb{Z}$ with an addition $+$, neutral: 0, inverse of n is $-n$, commutative,
- $\mathbb{N}$ with $+$ and 0 is not a group because there's no inverse for 1.
- $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ with product and 1.

Lemma 2

Let G be a group, then

1. there's a unique neutral element,
2. inverses are likewise unique.

Proof of Lemma 2.
1. Suppose that e_1, e_2 are neutral elements. Then $e_2 = e_2 * e_1 = e_1$.
2. Let $g \in G$ and suppose that h_1, h_2 are its inverses, then $(h_1 * g) * h_2 = h_1 * (g * h_2) \implies e * h_2 = h_1 * e$.

□

Notation:

We will use g^{-1} meaning the inverse of g in a group G. In Abelian groups with addition the inverse is denoted as $-g$.

1.1 Vector Spaces

Example 2

Recall $\mathbb{R}^2 = \{(x,y) | x, y \in \mathbb{R}\}$. We have a zero vector $(0,0)$ and an addition $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$. Easy to proof that it's a group.
But there's something more in it: $\forall \lambda \in \mathbb{R} : \lambda(x,y) = (\lambda x, \lambda y)$. Interaction with $+$: $\lambda((x,y) + (x',y')) = \lambda(x,y) + \lambda(x',y')$ and $(\lambda + \mu)(x,y) = \lambda(x,y) + \mu(x,y)$.

Notation:

From now on, $\mathbb{K}$ denotes $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$.

Definition 3: Vector Space

A $\mathbb{K}$-*vector space(vs)* is a set E together with the following structures:

- A stucture of a commutative group with $+$ and neutral element denoted 0,
- Multiplication by scalars in $\mathbb{K}$:

$$\cdot : \mathbb{K} \times E \longrightarrow E, (\lambda, (x, y)) \longmapsto \lambda(x, y).$$

  Satisfying

  - distributivity over $+_E$: $\lambda(u +_E v) = \lambda u +_E \lambda v$,
  - distributivity over addition in $\mathbb{K}$: $(\lambda + \mu)u = \lambda u +_E \lambda v$,
  - associativety $\forall \lambda, \mu \in \mathbb{K}, u \in E$: $(\lambda \cdot \mu) \cdot_E u = \lambda \cdot_E (\mu \cdot_E u)$,
  - identity/unit: $\forall u \in E : 1 \cdot u = u$.

Definition 4: Vector

Elements of a vector space are usually called *vectors*.

Example 3

- $E = \{0\}$ — zero vector space,
- $E = \mathbb{K}^n = \{(a_1, a_2, \ldots, a_n) \,|\, a_i \in \mathbb{K}\}$
  - addition: $(a_1, \ldots a_n) + (b_1, \ldots, b_n) = (a_1 + b_1, \ldots, a_n + b_n)$,
  - zero element: $(0, \ldots, 0)$ denoted $\vec{0}$,
  - inverse: $-(a_1, \ldots, a_n) = (-a_1, \ldots, -a_n)$,
  - multiplication by scalars: $\lambda(a_1, \ldots, a_n) = (\lambda a_1, \ldots, \lambda a_n)$.

  Note: $\mathbb{K}$ itself is a vector space over $\mathbb{K}$.
- Sequences $\mathbb{K}^{\mathbb{N}} = \{(a_n)_{n \leq 0} \,|\, a_i \in \mathbb{K}\}$,
- If X is a non-empty set and $f(X, \mathbb{K}) = \{f : X \longrightarrow \mathbb{K}\}$.
  - addition: $f, g \in f(X, \mathbb{K})$ we define $(f + g)(x) = f(x) + g(x)$,
  - zero element: $0(x) = 0$,
  - inverse: if $f \in f(X, \mathbb{K})$ then $(-f)(x) = -f(x)$.
  - multiplication by scalars if $\lambda \in \mathbb{K}$ and $f \in f(X, \mathbb{K})$ we define $(\lambda f)(x) = \lambda f(x)$.

  Note: $f(X, \mathbb{K})$ generalises previous examples:
  - $\mathbb{K}^n$ is the same as $f(\{1, \ldots, n\}, \mathbb{K})$.

Lemma 5

Let E be a vector space over $\mathbb{K}$. Then:

- $\forall v \in E : 0_{\mathbb{K}} \cdot v = 0_E$,
- $\forall \lambda \in \mathbb{K} : \lambda \cdot 0_E = 0_E$,
- $\forall v \in E : (-1) \cdot v = -v$,
- if $\lambda \in \mathbb{K}, u \in E$ then $\lambda u = 0_E \Leftrightarrow \lambda = 0_{\mathbb{K}}$ or $u = 0_E$.

Conventions:

- From now on, we use 0 instead of $0_{\mathbb{K}}$ and 0_E.
- In E, $u, v \in E$ we write $u - v$ meaning $u + (-v)$.
- We write $u + v + w$ instead of $u + (v + w)$ and $(u + v) + w$.
- We sometimes write λu instead of $\lambda \cdot u$.

More generally, if I is a finite(index) set and $\{u_i\}_{i \in I}$ is a collection of vectors then we can write $\sum_{i \in I} u_i$.

2 Lecture II: Vector subspaces

Example 4

In $\mathbb{K}^3$ we can consider $F = \{(x, y, 0) \mid x, y \in \mathbb{K}\}$.
We observe:

1. if $(x, y, z), (x', y', z') \in F$ then $(x + x', y + y', z + z') \in F$,
2. if $(x, y, z) \in F$ and $\lambda \in \mathbb{K}$ then $(\lambda x, \lambda y, \lambda z) \in F$,
3. $(0, 0, 0) \in F$.

Definition 6: Vector subspace

Let E be a $\mathbb{K}$-vector space and $F \subseteq E$. We say that F is a *vector subspace(vss)* of E if F inherits from E a structure of $\mathbb{K}$-vs.
Namely:

1. F is closed under addition in E,
2. F is closed under multiplication by scalars,
3. $0 \in F$.

Example 5: Vector subspaces

- xy-plane in $\mathbb{R}^3$,
- $\{0\}$ is always a $\mathbb{K}$-vss of a $\mathbb{K}$-vs,
- $\emptyset$ is never a $\mathbb{K}$-vss,
- A $\mathbb{K}$-vs E is also a $\mathbb{K}$-vss of itself,
- $\mathbb{K}[X]$ — the set of all polynomials over $\mathbb{K}$ (expressions of the form $a_0+a_1X+\dots a_dX^d$ for some $d \geq 0$ (not fixed) and $a_i \in \mathbb{K}$). $\mathbb{K}[X]$ is a $\mathbb{K}$-vs and $\mathbb{K}[X]_{\leq d}$ is a $\mathbb{K}$-vss of $K[X]$.
- In $\mathbb{R}^2$, consider $F = \{(x, y) \in \mathbb{R}^2 \mid x \cdot y = 0\}$. F is not a $\mathbb{K}$-vss of $\mathbb{R}^2$ because $(1, 0) \in F$ and $(0, 1) \in F$, but $(1, 0) + (0, 1) = (1, 1) \notin F$.

2.1 Span of a Family of Vectors

Let E be a $\mathbb{K}$-vector space and $u_1, \dots, u_n \in E$.

Definition 7: Span

The *span* of $u_1, \dots, u_n$, denoted by $\text{Vect}(u_1, \dots, u_n)$ or $\langle u_1, \dots, u_n \rangle$, is defined as:

$$\langle u_1, \dots, u_n \rangle = \{\lambda_1 u_1 + \dots + \lambda_n u_n \mid \lambda_i \in \mathbb{K}\}.$$

Lemma 8

Let E and $u_1, \dots, u_n$ be as above. Then $\langle u_1, \dots, u_n \rangle$ is a $\mathbb{K}$-vss of E.

Generalisation:

If $\{u_i\}_{i \in I}$ is a family of vectors in E (possibly infinite), we define:

$$\langle \{u_i\}_{i \in I} \rangle = \text{finite linear combinations of vectors in } \{u_i\}_{i \in I} = \left\{ \sum_{j=1}^{k} \lambda_j u_{i_j} \;\middle|\; k \geq 0,\ \lambda_j \in \mathbb{K},\ i_j \in I \right\}.$$

Example 6

1. $\mathbb{K}^n = \langle (1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, 0, \dots, 1) \rangle$, since $(a_1, \dots, a_n) = a_1(1, 0, \dots, 0) + \dots + a_n(0, 0, \dots, 1)$.
2. $\mathbb{K}[X] = \langle 1, X, X^2, \dots \rangle$.
3. $\mathbb{K}[X]_{\leq d} = \langle 1, X, X^2, \dots, X^d \rangle$.

Remark 9

- There can be repeated elements in the family.
- $\langle \emptyset \rangle = \{0\}$.

- Shorthand notation: $\langle A \rangle$, where $A = \{u_i\}_{i \in I}$.

Lemma 10

Let the notation be as above.

1. $\langle \{u_i\}_{i \in I} \rangle$ is a vss of E containing $\{u_i\}_{i \in I}$.
2. If $H \subseteq E$ is a vss such that $\{u_i\}_{i \in I} \subseteq H$, then $\langle \{u_i\}_{i \in I} \rangle \subseteq H$.

Informally: $\langle \{u_i\}_{i \in I} \rangle$ is the smallest vss which contains $\{u_i\}_{i \in I}$.

Proof of Lemma 10. By contradiction. □

2.2 Intersection and Sum of Vector Subspaces

Proposition 11: Intersection of Subspaces

Let F, G be vector subspaces of a $\mathbb{K}$-vs E. Then $F \cap G = \{u \in E \mid u \in F \text{ and } u \in G\}$ is a vss of E.

PROOF.
- $0 \in F$ and $0 \in G \implies 0 \in F \cap G$.
- If $x, y \in F \cap G$, then $x, y \in F \implies x + y \in F$, and $x, y \in G \implies x + y \in G$, so $x + y \in F \cap G$.
- Similarly, $\lambda x \in F \cap G$ for any $\lambda \in \mathbb{K}$.

□

Remark 12

Never write $F \cap G = \emptyset$ (the zero element 0 is always in $F \cap G$).

Definition 13: Sum of Subspaces

Let E be a $\mathbb{K}$-vs and $F, G \subseteq E$ two vss. We define the sum of F and G by:

$$F + G = \{u + v \in E \mid u \in F, v \in G\}.$$

Lemma 14

1. $F + G$ is a vss of E.
2. $F, G \subseteq F + G$.
3. If $H \subseteq E$ is a vss such that $F \subseteq H$ and $G \subseteq H$, then $F + G \subseteq H$.

Proof of (1). $0 \in F$, $0 \in G$ and $0 + 0 = 0 \in F + G$, $u_1 + v_1 \in F + G, u_2 + v_2 \in F + G \implies (u_1 + u_2) + (v_1 + v_2) \in F + G$, Similarly for scalar multiplication. □

2.3 Direct Sums and Complementary Subspaces

Lemma 15

Let E be a $\mathbb{K}$-vs and $F, G \subseteq E$ be vss. The following are equivalent:

1. $F \cap G = \{0\}$.
2. $\forall u \in F + G$, there exist a unique $v \in F$ and a unique $w \in G$ such that $u = v + w$.

Proof of (1) $\implies$ (2). Assume $F \cap G = \{0\}$. Let $u \in F + G$ and suppose $u = v_1 + w_1 = v_2 + w_2$ with $v_1, v_2 \in F$ and $w_1, w_2 \in G$. Then $v_1 - v_2 = w_2 - w_1$. Since $v_1 - v_2 \in F$ and $w_2 - w_1 \in G$, we have $v_1 - v_2 \in F \cap G = \{0\}$. Thus $v_1 - v_2 = 0 \implies v_1 = v_2$, and similarly $w_1 - w_2 = 0 \implies w_1 = w_2$. □

Definition 16: Direct Sum

If any of the equivalent conditions of Lemma 15 is satisfied, we say that F and G form a *direct sum*. We write $F \oplus G$ for $F + G$ in this setting.

Definition 17: Complementary Subspace

Let $F \subseteq E$ be a vss of E. A *complement* for F is a vss $G \subseteq E$ such that $F \oplus G = E$. (Hence $F + G = E$ and $F \cap G = \{0\}$). In such case, we say that F and G are complementary.

Example 7

In $\mathbb{R}^3$, the xy-plane and the z-axis are complementary subspaces.

Proposition 18: Existence of Complement

Let E be a $\mathbb{K}$-vs and $F \subseteq E$ a $\mathbb{K}$-vss. Then there exists a complement for F.

3 Lecture III: Finite Dimensional Vector Spaces

Definition 19: Finite Dimensionality

Let E be a $\mathbb{K}$-vs. We say E is *finite dimensional* if there exist $e_1, \ldots, e_n \in E$ such that $E = \langle e_1, \ldots, e_n \rangle$. In this case, we define:

$$\dim E = \min\{n \geq 0 \mid \exists e_1, \ldots, e_n \in E \text{ s.t. } \langle e_1, \ldots, e_n \rangle = E\}.$$

Remark 20

$\dim E = 0 \iff E = \{0\}$. If E does not have finite dimension, we say E is *infinite dimensional* and write $\dim E = +\infty$.

Definition 21: Linear Independence

Let $u_1, \dots, u_n \in E$. We say they are *linearly independent* if:

$$\lambda_1 u_1 + \cdots + \lambda_n u_n = 0 \implies \lambda_1 = \cdots = \lambda_n = 0 \quad (\lambda_i \in \mathbb{K}).$$

More generally, a family $\{u_i\}_{i \in I}$ is linearly independent if any finite subfamily is linearly independent.

Lemma 22

Let E be a $\mathbb{K}$-vs, and $u_1, \dots, u_n \in E$. The following are equivalent:

1. $u_1, \dots, u_n$ are linearly independent.
2. $\forall u \in \langle u_1, \dots, u_n \rangle$, there exist unique $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that $u = \lambda_1 u_1 + \cdots + \lambda_n u_n$.

Lemma 23

Suppose $\{u_i\}_{i \in I}$ is a linearly independent family. Then any subfamily $\{u_j\}_{j \in J}$ (with $J \subseteq I$) is linearly independent.

Proposition 24

Let E be a finite dimensional vs with $\dim E = n$. Assume $E = \langle e_1, \dots, e_n \rangle$. Then $e_1, \dots, e_n$ are linearly independent.

Definition 25: Linear Dependence

If vectors are not linearly independent, we say they are *linearly dependent*.

Lemma 26

A family $u_1, \dots, u_n \in E$ is linearly dependent if and only if there exists some $i \in \{1, \dots, n\}$ such that $u_i \in \langle u_1, \dots, u_{i-1}, u_{i+1}, \dots, u_n \rangle$.

3.1 Bases

Definition 27: Basis

Let E be a $\mathbb{K}$-vs. A *basis* of E is a family of vectors $\{e_i\}_{i \in I}$ such that:

1. $E = \langle \{e_i\}_{i \in I} \rangle$ (spanning family),
2. $\{e_i\}_{i \in I}$ is linearly independent.

Remark 28

1. If $e_1, \dots, e_n$ is a basis, then for all $u \in E$, there exist unique $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that $u = \sum_{i=1}^{n} \lambda_i e_i$.
2. By convention, if $E = \{0\}$, then $\emptyset$ is a basis of E.

Theorem 29: Existence of Basis

Every vector space admits a basis.

Theorem 30: Dimension Theorem

Let E be a finite dimensional space with $\dim E = n$. Then all bases of E have exactly n vectors.

Lemma 31

Let $u_1, \dots, u_n \in E$ be linearly independent vectors in a vs E. Let $k \in \{1, \dots, n-1\}$. Then:

$$\langle u_1, \dots, u_k \rangle \cap \langle u_{k+1}, \dots, u_n \rangle = \{0\}.$$

Lemma 32

Let $u_1, \dots, u_n \in E$. Let $k \in \{1, \dots, n\}$ and $v_{k+1}, \dots, v_n \in \langle u_1, \dots, u_k \rangle$. Then:

$$\langle u_1, \dots, u_n \rangle = \langle u_1, \dots, u_k, u_{k+1} + v_{k+1}, \dots, u_n + v_n \rangle.$$

Lemma 33

In the setting of Lemma 32, assume moreover that $u_1, \dots, u_n$ are linearly independent. Then $u_1, \dots, u_k, u_{k+1} + v_{k+1}, \dots, u_n + v_n$ are also linearly independent.

Proposition 34: Steinitz Exchange Lemma / Comparison

Let E be a $\mathbb{K}$-vs. Let $u_1, \dots, u_m \in E$ be a linearly independent family, and $e_1, \dots, e_n \in E$ another family. Suppose $\langle u_1, \dots, u_m \rangle \subseteq \langle e_1, \dots, e_n \rangle$. Then $m \leq n$, and equality holds if and only if $\langle u_1, \dots, u_m \rangle = \langle e_1, \dots, e_n \rangle$.

Proof Sketch (Gaussian Elimination Idea). First, since $u_1 \in \langle e_1, \dots, e_n \rangle$, write $u_1 = a_{11}e_1 + \cdots + a_{n1}e_n$. Since $u_1 \neq 0$, at least one coefficient is non-zero (say $a_{11} \neq 0$). We replace e_1 with u_1. Next, eliminate the component along u_1 from u_2 by setting $u_2' = u_2 - \frac{b_1}{a_{11}} u_1$. By Lemma 32 and Lemma 33, the new family remains linearly independent and spans the same subspace. Proceeding inductively yields $m \leq n$. □

Corollary 35

Let E be a $\mathbb{K}$-vs and $e_1, \dots, e_n \in E$ linearly independent. Then $e_1, \dots, e_n$ is a basis of $\langle e_1, \dots, e_n \rangle$, so $\dim \langle e_1, \dots, e_n \rangle = n$.

Corollary 36

Let E be a finite dimensional vs with $\dim E = n$. Let $u_1, \dots, u_n \in E$ be linearly independent. Then $E = \langle u_1, \dots, u_n \rangle$, so $u_1, \dots, u_n$ forms a basis of E.

Corollary 37

Let E be a finite dimensional vs and $F \subseteq E$ a vss. Then F is finite dimensional and $\dim F \leq \dim E$. Moreover, $\dim F = \dim E \iff F = E$.

Example 8

$\dim \mathbb{K}^n = n$, since $\mathbb{K}^n = \langle (1, 0, \dots, 0), (0, 1, \dots, 0), \dots, (0, \dots, 0, 1) \rangle$.

3.2 Extraction and Completion of Bases

Proposition 38: Extraction of Bases

Let E be a $\mathbb{K}$-vs. Assume $E = \langle e_1, \dots, e_n \rangle$. Then one can find a subfamily $e_{i_1}, \dots, e_{i_k}$ which is a basis of E.

Proposition 39: Incomplete Basis Theorem

Let E be a finite dimensional vs. Let $e_1, \dots, e_m \in E$ be linearly independent. Then $e_1, \dots, e_m$ can be completed into a basis of E, i.e., there exist $e_{m+1}, \dots, e_n$ such that $e_1, \dots, e_n$ is a basis of E.

Corollary 40: Existence of Complement (Finite Dim)

Let E be a finite dimensional vs, and $F \subseteq E$ a vss. Then there exists a vss $G \subseteq E$ such that $E = F \oplus G$.

Theorem 41: Grassmann's Formula

Let E be a finite dimensional vs, and $F, G \subseteq E$ two vss. Then:

$$\dim(F + G) = \dim F + \dim G - \dim(F \cap G).$$

In particular, $\dim(F + G) \leq \dim F + \dim G$, with equality if and only if $F \cap G = \{0\}$.

Example 9

Let $E = \mathbb{R}[X]_{\leq 3} = \{P \in \mathbb{R}[X] \mid \deg P \leq 3\}$, so $\dim E = 4$. Consider $F = \{P \in E \mid P(0) = P(1) = 0\}$ and $G = \{P \in E \mid P(2) = P(3) = 0\}$.

- $F \cap G = \{0\}$: If $P \in F \cap G$, then P has 4 distinct roots $(0, 1, 2, 3)$, but $\deg P \leq 3 \implies P = 0$.
- $\dim F \geq 2$ since $X(X-1), X^2(X-1) \in F$ are linearly independent. Similarly

$\dim G \geq 2$.

- By Grassmann's formula: $\dim(F+G) = \dim F + \dim G \geq 2+2 = 4 = \dim E \implies E = F \oplus G$.

4 Lecture IV: Linear Maps

Definition 42: Linear Map

Let E, F be two $\mathbb{K}$-vs. A map $f : E \to F$ is $\mathbb{K}$-*linear* if:

1. $\forall u, v \in E : f(u+v) = f(u) + f(v)$,
2. $\forall \lambda \in \mathbb{K}, u \in E : f(\lambda u) = \lambda f(u)$.

Equivalently, $\forall \lambda, \mu \in \mathbb{K}, u, v \in E : f(\lambda u + \mu v) = \lambda f(u) + \mu f(v)$.

Example 10

1. Zero map $0 : E \to F, u \mapsto 0$.
2. Identity map $\mathrm{Id}_E : E \to E, u \mapsto u$.
3. Derivative map $D : \mathbb{K}[X] \to \mathbb{K}[X], P(X) \mapsto P'(X)$.
4. Evaluation map: Fix $x_0 \in X$. $\mathrm{ev}_{x_0} : \mathcal{F}(X, \mathbb{R}) \to \mathbb{R}, f \mapsto f(x_0)$.

Lemma 43

Let $f : E \to F$ be a linear map. Then $f(0_E) = 0_F$.

Notation:

We denote by $\mathrm{Hom}_{\mathbb{K}}(E, F)$ or $\mathcal{L}(E, F)$ the set of all $\mathbb{K}$-linear maps from E to F. If $E = F$, we denote $\mathrm{Hom}_{\mathbb{K}}(E, E) = \mathrm{End}(E)$, and elements of $\mathrm{End}(E)$ are called *endomorphisms* of E.

Proposition 44: Properties of Linear Maps

1. $\mathrm{Hom}_{\mathbb{K}}(E, F)$ is a $\mathbb{K}$-vector space under point-wise addition and scalar multiplication.
2. If $f \in \mathrm{Hom}_{\mathbb{K}}(E, F)$ and $g \in \mathrm{Hom}_{\mathbb{K}}(F, G)$, then the composition $g \circ f : E \to G$ is a $\mathbb{K}$-linear map.

Proposition 45: Images of Subspaces

Let $f : E \to F$ be linear.

1. If $E' \subseteq E$ is a vss of E, then $f(E') = \{f(u) \mid u \in E'\}$ is a vss of F.
2. If $F' \subseteq F$ is a vss of F, then $f^{-1}(F') = \{u \in E \mid f(u) \in F'\}$ is a vss of E.

Definition 46: Image and Kernel

Let $f : E \to F$ be linear.

- The *image* of f is $\mathrm{Im}(f) = f(E) \subseteq F$, which is a vss of F.
- The *kernel* of f is $\mathrm{Ker}(f) = f^{-1}(\{0\}) = \{u \in E \mid f(u) = 0\}$, which is a vss of E.

Proposition 47: Injectivity and Surjectivity

1. f is surjective $\iff \mathrm{Im}(f) = F$.
2. f is injective $\iff \mathrm{Ker}(f) = \{0\}$.

Definition 48: Isomorphism and Automorphism

A linear map $f : E \to F$ is an *isomorphism* if it is bijective. An isomorphism $f : E \to E$ is called an *automorphism* of E.

Proposition 49

If $f : E \to F$ is an isomorphism, then its inverse $f^{-1} : F \to E$ is also a $\mathbb{K}$-linear map.

Proposition 50: Linear Map Determination by a Basis

Let E, F be vector spaces. Fix a basis $\{e_i\}_{i \in I}$ of E and a collection of vectors $\{f_i\}_{i \in I} \subseteq F$. Then there exists a unique linear map $f : E \to F$ such that $f(e_i) = f_i$ for all $i \in I$.

Proposition 51

Let $f : E \to F$ be a linear map.

1. f is injective $\iff$ f preserves linear independence.
2. f is surjective $\iff$ f sends any spanning set to a spanning set of F.
3. f is bijective $\iff$ f maps any basis of E to a basis of F.

Corollary 52: Dimension and Linear Maps

Let E, F be finite dimensional $\mathbb{K}$-vs, and $f : E \to F$ linear.

1. If f is injective, then $\dim E \leq \dim F$.
2. If f is surjective, then $\dim E \geq \dim F$.
3. If f is bijective, then $\dim E = \dim F$.

Corollary 53: Equal Dimension Criteria

Let E, F be finite dimensional vector spaces with $\dim E = \dim F$. Let $f : E \to F$ be a linear map. The following are equivalent:

1. f is injective.
2. f is surjective.
3. f is bijective.

4.1 Rank and Nullity

Definition 54: Rank and Nullity

Let $f : E \to F$ be a linear map.

1. The *rank* of f is $\mathrm{rk}(f) = \dim(\mathrm{Im}(f))$.
2. The *nullity* of f is $\mathrm{null}(f) = \dim(\mathrm{Ker}(f))$.

Theorem 55: Rank-Nullity Theorem

Let E, F be vector spaces with E finite dimensional, and $f : E \to F$ a linear map. Then:

$$\dim E = \mathrm{rk}(f) + \mathrm{null}(f).$$

5 Lecture V: Matrices and Determinants

Let $A \in \mathrm{Mat}_{m\times n}(\mathbb{K})$ be an $m \times n$ matrix. We associate to A the linear map $f_A : \mathbb{K}^n \to \mathbb{K}^m, X \mapsto AX$.

Definition 56: Rank and Nullity of a Matrix

$\mathrm{rk}(A) = \mathrm{rk}(f_A)$ and $\mathrm{null}(A) = \mathrm{null}(f_A)$.

Proposition 57: Matrix Representation

Let E, F be finite dimensional vs with basis $\mathcal{B} = (e_1, \dots, e_n)$ for E and $\mathcal{B}' = (u_1, \dots, u_m)$ for F. For $f \in \mathrm{Hom}_{\mathbb{K}}(E, F)$, the matrix of f with respect to $\mathcal{B}$ and $\mathcal{B}'$, denoted $\mathrm{Mat}_{\mathcal{B},\mathcal{B}'}(f)$ or M_f, is defined by:

$$f(e_j) = \sum_{i=1}^{m} a_{ij} u_i \quad (j = 1, \dots, n), \qquad M_f = (a_{ij})_{m\times n}.$$

Corollary 58

$\dim \mathrm{Hom}_{\mathbb{K}}(E, F) = \dim \mathrm{Mat}_{m\times n}(\mathbb{K}) = m \cdot n$.

Definition 59: Invertibility of Matrix

A matrix $A \in \mathrm{Mat}_n(\mathbb{K})$ is invertible if its associated linear map f_A is bijective. In this case, A^{-1} satisfies $AA^{-1} = A^{-1}A = \mathrm{I}_n$.

Lemma 60: Equivalences of Invertibility

Let $A \in \mathrm{Mat}_n(\mathbb{K})$. The following are equivalent:

1. A is invertible.
2. $\mathrm{rk}(A) = n$.
3. $\mathrm{null}(A) = 0$.
4. $\exists B$ s.t. $AB = \mathrm{I}_n$.
5. $\exists B$ s.t. $BA = \mathrm{I}_n$.
6. $AX = Y$ has a unique solution X for every $Y \in \mathbb{K}^n$.

5.1 Preliminaries on Permutations

Definition 61: Permutation

Let S be a non-empty set. A *permutation* on S is a bijective map $f : S \to S$.

Lemma 62

The set of permutations of a non-empty set S, endowed with the composition of maps, has the structure of a group.

Definition 63: Symmetric Group

The *symmetric group of order* n is the set of permutations on $\{1, 2, \ldots, n\}$ (for $n \geq 1$). It is denoted by S_n or $\mathfrak{S}_n$.

Notation:

If $\sigma \in S_n$, we write:

$$\sigma = \begin{pmatrix} 1 & 2 & \ldots & n \\ \sigma(1) & \sigma(2) & \ldots & \sigma(n) \end{pmatrix}$$

Lemma 64

1. $\#S_n = n!$
2. If $n \geq 3$, S_n is not commutative (non-Abelian).

5.2 Cycles and Transpositions

Definition 65: Cycles and Transpositions

1. A *cycle* of S_n (or cyclic permutation) is a permutation $\sigma \in S_n$ such that there exist distinct elements $i_1, i_2, \ldots, i_k \in \{1, 2, \ldots, n\}$ with:
$$\sigma(i_1) = i_2, \quad \sigma(i_2) = i_3, \quad \ldots, \quad \sigma(i_k) = i_1,$$
and $\sigma(j) = j$ for all $j \notin \{i_1, \ldots, i_k\}$. We denote such a cycle by $(i_1\, i_2\, \ldots\, i_k)$.
2. If σ is a cycle as in (1), we call $\{i_1, \ldots, i_k\}$ the *orbit* of σ.
3. A *transposition* is a cycle of length 2, i.e., of the form $\tau = (i\ j)$ with $i \neq j$.

Proposition 66: Decomposition of Permutations

1. Any permutation can be written as a composition of cycles with disjoint orbits. These cycles commute with each other, and the decomposition is unique up to order: $\sigma = \sigma_1 \circ \sigma_2 \circ \cdots \circ \sigma_m$.
2. Any cyclic permutation is a composition of transpositions.
3. Any permutation is a composition of transpositions. This decomposition is not unique, but the number of transpositions always has a well-defined parity (even or odd).

5.3 Parity and Signature

Definition 67: Parity and Signature

The *parity* of a permutation $\sigma \in S_n$ is the parity of the number of transpositions in any of its decompositions. The *signature* of $\sigma \in S_n$ is defined by:
$$\epsilon(\sigma) = \begin{cases} +1 & \text{if } \sigma \text{ is even,} \\ -1 & \text{if } \sigma \text{ is odd.} \end{cases}$$

Lemma 68

The map $\epsilon : S_n \to \{\pm 1\}$ satisfies:

1. $\epsilon(\sigma_1 \circ \sigma_2) = \epsilon(\sigma_1)\epsilon(\sigma_2)$,
2. $\epsilon(\mathrm{id}) = 1$.

Alternative Definition via Inversions:

Let $\sigma \in S_n$. An *inversion* in σ is a pair (i, j) with $1 \leq i < j \leq n$ such that $\sigma(i) > \sigma(j)$. If $I(\sigma)$ is the total number of inversions in σ, then:
$$\epsilon(\sigma) = (-1)^{I(\sigma)}$$

5.4 Alternating n-Forms

Definition 69: Alternating n-Form

Let E be a $\mathbb{K}$-vs and $n \geq 1$ an integer. An *alternating n-form* is a map:

$$\Theta : \underbrace{E \times \cdots \times E}_{n \text{ times}} \longrightarrow \mathbb{K}, \quad (e_1, \ldots, e_n) \longmapsto \Theta(e_1, \ldots, e_n) \in \mathbb{K}$$

such that:

1. Θ is linear in every variable (multilinear),
2. If $e_i = e_j$ for some $i < j$, then $\Theta(e_1, \ldots, e_i, \ldots, e_j, \ldots, e_n) = 0$ (alternating property).

The set of alternating n-forms on E is denoted by $\mathcal{A}_n(E)$.

Lemma 70

Let $\Theta \in \mathcal{A}_n(E)$ and $\sigma \in S_n$. Then:

$$\Theta(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \epsilon(\sigma)\Theta(e_1, \ldots, e_n).$$

Proof. Since any permutation $\sigma \in S_n$ can be decomposed into a product of k transpositions, it suffices to prove the statement for a single transposition $\tau = (i\ j)$ with $i < j$. Consider the vectors $x, y \in E$ placed at the i-th and j-th positions. By the alternating property:

$$\Theta(\ldots, x + y, \ldots, x + y, \ldots) = 0.$$

Expanding by multilinearity yields:

$$\Theta(\ldots, x, \ldots, x, \ldots) + \Theta(\ldots, x, \ldots, y, \ldots) + \Theta(\ldots, y, \ldots, x, \ldots) + \Theta(\ldots, y, \ldots, y, \ldots) = 0.$$

Since terms with repeated arguments vanish, this simplifies to:

$$\Theta(\ldots, x, \ldots, y, \ldots) + \Theta(\ldots, y, \ldots, x, \ldots) = 0 \implies \Theta(e_{\tau(1)}, \ldots, e_{\tau(n)}) = -\Theta(e_1, \ldots, e_n).$$

Since $\epsilon(\tau) = -1$, the result holds for transpositions. By induction on the number of transpositions k, we obtain $\Theta(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = (-1)^k \Theta(e_1, \ldots, e_n) = \epsilon(\sigma)\Theta(e_1, \ldots, e_n)$. □

Theorem 71: Determinant Form

Let E be a finite dimensional $\mathbb{K}$-vs with $\dim E = n \geq 1$. Then:

1. $\dim \mathcal{A}_n(E) = 1$.
2. If $\mathcal{B} = (e_1, \ldots, e_n)$ is a basis of E, then there exists a unique associated basis element of $\mathcal{A}_n(E)$, denoted by $\det_{\mathcal{B}}$, characterized by:

$$\det_{\mathcal{B}}(e_1, \ldots, e_n) = 1.$$

Concretely, if $(u_1, \ldots, u_n) \in E^n$ with $u_j = \sum_{i=1}^n a_{ij} e_i$, then:

$$\det_{\mathcal{B}}(u_1, \ldots, u_n) = \sum_{\sigma \in S_n} \epsilon(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \ldots a_{\sigma(n),n} \quad (*).$$

PROOF. Let $\Theta \in \mathcal{A}_n(E)$. Expanding each $u_j = \sum_{i_j=1}^n a_{i_j,j} e_{i_j}$ by multilinearity gives:

$$\Theta(u_1, \ldots, u_n) = \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n a_{i_1,1} \ldots a_{i_n,n}\, \Theta(e_{i_1}, \ldots, e_{i_n}).$$

By the alternating property, $\Theta(e_{i_1}, \ldots, e_{i_n}) = 0$ whenever any index is repeated. Thus, non-zero terms occur only when $(i_1, \ldots, i_n)$ is a permutation of $(1, \ldots, n)$, i.e., $i_j = \sigma(j)$ for some $\sigma \in S_n$. Using Lemma 70:

$$\Theta(u_1, \ldots, u_n) = \sum_{\sigma \in S_n} a_{\sigma(1),1} \ldots a_{\sigma(n),n}\, \Theta(e_{\sigma(1)}, \ldots, e_{\sigma(n)}) = \left(\sum_{\sigma \in S_n} \epsilon(\sigma) a_{\sigma(1),1} \ldots a_{\sigma(n),n} \right) \Theta(e_1, \ldots, e_n).$$

Setting $\Theta(e_1, \ldots, e_n) = 1$ uniquely defines $\det_{\mathcal{B}}$, proving that $\mathcal{A}_n(E)$ is 1-dimensional and spanned by $\det_{\mathcal{B}}$. □

Definition 72: Determinant of Matrices

1. For $E = \mathbb{K}^n$, we simply denote by det the alternating n-form $\det_{\mathcal{B}}$, where $\mathcal{B} = (e_1, \ldots, e_n)$ is the standard basis.
2. If $A \in \mathrm{Mat}_n(\mathbb{K})$ with columns $(C_1, \ldots, C_n)$, then we define:

$$\det A = \det(C_1, \ldots, C_n).$$

If $A = (a_{ij})_{1 \le i,j \le n}$, then $\det A$ is given by formula (*).

5.5 Properties of Determinants

Focusing on $\det A$ for $A \in \mathrm{Mat}_n(\mathbb{K})$ or $\det(C_1, \ldots, C_n)$:

1. **Multilinearity:**

$$\det\left(C_1, \ldots, \sum_{k=1}^m \lambda_k C_i^{(k)}, \ldots, C_n\right) = \sum_{k=1}^m \lambda_k \det\left(C_1, \ldots, C_i^{(k)}, \ldots, C_n\right).$$

In particular, scaling all n columns yields $\det(\lambda A) = \lambda^n \det(A)$.

2. **Alternating Property:** If $i \neq j$, then:

$$\det(\ldots, C_i, \ldots, C_j, \ldots) = -\det(\ldots, C_j, \ldots, C_i, \ldots).$$

In particular, if $C_i = C_j$, then $\det(C_1, \ldots, C_n) = 0$.

3. **Invariance under Column Operations:** If $C_i' = C_i + \sum_{j \neq i} \lambda_j C_j$, then by multilinearity and alternating properties:

$$\det(C_1, \ldots, C_i', \ldots, C_n) = \det(C_1, \ldots, C_i, \ldots, C_n).$$

Note: Properties (1) to (3) allow simplifying computations of determinants using Gaussian elimination, introducing a sign change whenever two columns are swapped.

4. **Transpose:** If $A = (a_{ij})_{i,j}$, define ${}^tA = (a_{ji})_{i,j}$ (the transpose of A). Then:

$$\det A = \det\left({}^tA\right).$$

Proof Idea: Every permutation σ has $\epsilon(\sigma) = \epsilon(\sigma^{-1})$, reindexing the product $\prod a_{\sigma(i),i}$ over $\tau = \sigma^{-1}$ yields the sum for ${}^tA$. *Corollary:* Properties (1) to (3) hold identically for rows.

5. **Multiplicativity:**
$$\det(AB) = \det(A)\det(B).$$

6. **Invertibility Criterion:** $C_1, \dots, C_n$ are linearly independent $\iff \det(C_1, \dots, C_n) \neq 0$. Equivalently, A is invertible $\iff \det(A) \neq 0$. In this case:
$$\det\left(A^{-1}\right) = \det(A)^{-1} = \frac{1}{\det(A)}.$$

5.6 Inductive Rule and Inverse Matrix

Let $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{K})$. Define A_{ij} as the matrix obtained from A by deleting the i-th row and j-th column. Define the cofactor:
$$a^*_{ij} = (-1)^{i+j} \det(A_{ij}).$$

Laplace Expansion:

For every fixed column j:
$$\det A = a_{1j}a^*_{1j} + a_{2j}a^*_{2j} + \cdots + a_{nj}a^*_{nj}.$$

Since $\det({}^tA) = \det A$, we have a similar expansion along any row i:
$$\det A = a_{i1}a^*_{i1} + a_{i2}a^*_{i2} + \cdots + a_{in}a^*_{in}.$$

Definition 73: Comatrix

Let $A \in \mathrm{Mat}_n(\mathbb{K})$. We define the *comatrix* (or matrix of cofactors) of A as:
$$\mathrm{com}(A) = (a^*_{ij})_{1 \leq i,j \leq n}.$$

Proposition 74: Comatrix Inverse Formula

For any $A \in \mathrm{Mat}_n(\mathbb{K})$:
$$A \cdot {}^t\mathrm{com}(A) = {}^t\mathrm{com}(A) \cdot A = (\det A) \cdot \mathrm{I}_n.$$

In particular, if $\det A \neq 0$, then:
$$A^{-1} = \frac{1}{\det A}\,{}^t\mathrm{com}(A).$$

Proof. Consider the (i,k)-th entry of the matrix product $A \cdot {}^t\mathrm{com}(A)$:
$$\left(A \cdot {}^t\mathrm{com}(A)\right)_{ik} = \sum_{j=1}^{n} a_{ij}({}^t\mathrm{com}(A))_{jk} = \sum_{j=1}^{n} a_{ij}a^*_{kj}.$$

- If $i = k$, this is precisely the Laplace expansion of $\det A$ along the i-th row.
- If $i \neq k$, this sum corresponds to the Laplace expansion along row k of a modified matrix A' where row k is replaced by a copy of row i. Since A' has two identical rows, $\det A' = 0$.

Hence, $\left(A \cdot {}^t\mathrm{com}(A)\right)_{ik} = \delta_{ik} \det A$, so $A \cdot {}^t\mathrm{com}(A) = (\det A)\mathrm{I}_n$. Dividing by $\det A$ yields A^{-1}. □

Example 11

Compute A^{-1} for $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix}$:

$$\det A = 1 \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} - 0 \cdot \begin{vmatrix} -1 & 0 \\ 2 & 1 \end{vmatrix} + 1 \cdot \begin{vmatrix} -1 & 0 \\ 1 & 1 \end{vmatrix} = (1-2) + 0 + (-1-0) = -2.$$

Computing cofactors yields:

$$\operatorname{com}(A) = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -3 \\ -1 & -1 & 1 \end{pmatrix} \implies {}^t\operatorname{com}(A) = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -3 & 1 \end{pmatrix}.$$

Hence:

$$A^{-1} = -\frac{1}{2} \begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -3 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & -1/2 & 1/2 \\ 1/2 & 3/2 & -1/2 \end{pmatrix}.$$

5.7 Applications of Determinants

1. Rank of a Matrix via Minors

Let $A \in \operatorname{Mat}_{m\times n}(\mathbb{K})$. Recall that $\operatorname{rk}(A) = \operatorname{rk}(f_A)$, where $f_A : \mathbb{K}^n \to \mathbb{K}^m$. Note that $\operatorname{rk}(f) \leq \min(\dim E, \dim F)$.

Definition 75: Minor

Let $r \in \{1, 2, \ldots, \min(n, m)\}$. A *minor of order* r is the determinant of an $r \times r$ submatrix of A. If $A = (a_{ij})$, a minor is of the form:

$$\begin{vmatrix} a_{i_1 j_1} & \cdots & a_{i_1 j_r} \\ \vdots & \ddots & \vdots \\ a_{i_r j_1} & \cdots & a_{i_r j_r} \end{vmatrix} \quad \text{where } 1 \leq i_1 < \cdots < i_r \leq m \text{ and } 1 \leq j_1 < \cdots < j_r \leq n.$$

Proposition 76: Rank Characterization by Minors

Let $r = \operatorname{rk}(A)$. Then:

1. For any $s > r$ with $s \leq \min(n, m)$, all minors of order s vanish (equal zero).
2. There exists at least one minor of order r which is non-zero.

Hence, $\operatorname{rk}(A)$ is the largest integer $r \leq \min(n, m)$ such that there exists a non-zero minor of order r.

Corollary 77

$\operatorname{rk}(A) = \operatorname{rk}({}^tA)$.

2. Solving Systems of Linear Equations

Consider a system of linear equations:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = y_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = y_m \end{cases} \iff AX = Y, \text{ where } A = (a_{ij}),\ X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},\ Y = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}.$$

Three main cases arise:

1. A is injective ($\operatorname{rk}(A) = n \implies n \leq m$): there exists at most one solution for $AX = Y$.
2. A is surjective ($\operatorname{rk}(A) = m \implies m \leq n$): for every $Y \in \mathbb{K}^m$, there exists at least one solution.
3. A is invertible ($\operatorname{rk}(A) = n = m$): there exists a unique solution $X = A^{-1}Y$ for every $Y \in \mathbb{K}^n$.

Remark 78

If Y is given and $X_0, X_1 \in \mathbb{K}^n$ satisfy $AX_0 = Y$ and $AX_1 = Y$, then $A(X_0 - X_1) = 0 \implies X_0 - X_1 \in \operatorname{Ker}(A)$. Therefore, $\{X \mid AX = Y\} = X_0 + \operatorname{Ker}(A)$.

Proposition 79: Cramer's Rule

Suppose $A \in \operatorname{Mat}_n(\mathbb{K})$ is invertible. Let $Y \in \mathbb{K}^n$ be given, and let $X \in \mathbb{K}^n$ be the unique solution to $AX = Y$. Writing $A = (C_1, \ldots, C_n)$ in terms of its column vectors:

$$x_i = \frac{\det(C_1, \ldots, C_{i-1}, Y, C_{i+1}, \ldots, C_n)}{\det(C_1, \ldots, C_n)}, \qquad \forall i \in \{1, \ldots, n\}.$$

Proof. Since $AX = Y$, we can express Y as a linear combination of the columns of A:

$$Y = \sum_{j=1}^{n} x_j C_j = x_1 C_1 + \cdots + x_i C_i + \cdots + x_n C_n.$$

Substituting Y into the i-th column of the determinant matrix yields:

$$\det\left(C_1, \ldots, \underbrace{\sum_{j=1}^{n} x_j C_j}_{i\text{-th pos}}, \ldots, C_n\right) = \sum_{j=1}^{n} x_j \det\left(C_1, \ldots, \underbrace{C_j}_{i\text{-th pos}}, \ldots, C_n\right).$$

For $j \neq i$, the matrix inside the determinant has two identical columns (C_j at both the j-th and i-th positions), so those terms vanish. The only non-zero term is $j = i$:

$$\det(C_1, \ldots, Y, \ldots, C_n) = x_i \det(C_1, \ldots, C_i, \ldots, C_n) = x_i \det A.$$

Dividing by $\det A \neq 0$ completes the proof. □

6 Lecture VI: The Dual Vector Space

6.1 The Dual Space and Dual Bases

Definition 80: Dual Vector Space

Let E be a vector space over $\mathbb{K}$.

1. A *linear form on* E is a linear map $\omega : E \to \mathbb{K}$.
2. The *dual vector space of* E, denoted by E^* or $E^\vee$, is the vector space of all linear forms on E. Namely, $E^* = \mathcal{L}(E, \mathbb{K}) = \mathrm{Hom}_{\mathbb{K}}(E, \mathbb{K})$.

Example 12: Linear Forms

1. Let $u = (u_1, \ldots, u_n) \in \mathbb{K}^n$. We associate to it a linear form:

$$\mathbb{K}^n \longrightarrow \mathbb{K}, \quad \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \longmapsto u_1 x_1 + \cdots + u_n x_n.$$

   Note that this is the linear map associated to u seen as a row matrix in $\mathrm{Mat}_{1\times n}(\mathbb{K})$.

2. For a fixed scalar $a \in \mathbb{K}$, evaluation at a defines a linear form on polynomials:

$$\mathrm{ev}_a : \mathbb{K}[X] \longrightarrow \mathbb{K}, \quad P(X) \longmapsto P(a).$$

3. Let E be a vector space and $e \in E$ a non-zero vector. Let F be a complement for the span $\langle e \rangle$, so $E = \langle e \rangle \oplus F$. Projection onto $\langle e \rangle$ gives a linear map $p : E \to \langle e \rangle$. Choosing e as a basis for $\langle e \rangle$ gives an isomorphism $\iota : \langle e \rangle \to \mathbb{K}$ mapping $\lambda e \mapsto \lambda$. Composing $\omega = \iota \circ p : E \to \mathbb{K}$ yields a linear form: if $u \in E$ is uniquely written as $u = \lambda e + v$ with $v \in F$, then $\omega(u) = \lambda$.

Definition 81: Dual Basis Construction

Let E be a finite dimensional vector space with basis $\mathcal{B} = (e_1, \ldots, e_n)$. For every e_i, there exists a unique linear map $e_i^* : E \to \mathbb{K}$ defined by:

$$e_i^*(e_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Concretely, $e_i^*(a_1 e_1 + \cdots + a_n e_n) = a_i$ extracts the i-th coordinate of a vector in basis $\mathcal{B}$. The family $\mathcal{B}^* = (e_1^*, \ldots, e_n^*)$ is called the *dual basis* of $\mathcal{B}$.

Proposition 82: Properties of Dual Bases

Let E be a finite dimensional vector space.

1. If $e_1, \ldots, e_n$ is a basis of E, then $e_1^*, \ldots, e_n^*$ is a basis of E^*. In particular, $\dim E^* = \dim E$.

2. If $\omega : E \to \mathbb{K}$ is a linear form, then its expansion in the dual basis is:

$$\omega = \sum_{i=1}^{n} \omega(e_i) e_i^*.$$

Thus, the coefficients of ω in basis $(e_1^*, \ldots, e_n^*)$ are $(\omega(e_1), \ldots, \omega(e_n))$.

PROOF. Let $\omega \in E^*$. Consider the form $\omega' = \sum_{i=1}^{n} \omega(e_i) e_i^*$. Evaluating ω' on any basis vector e_j:

$$\omega'(e_j) = \sum_{i=1}^{n} \omega(e_i) e_i^*(e_j) = \sum_{i=1}^{n} \omega(e_i) \delta_{ij} = \omega(e_j).$$

Since two linear maps agreeing on a basis are equal, $\omega = \sum_{i=1}^{n} \omega(e_i) e_i^*$. This proves that $e_1^*, \ldots, e_n^*$ span E^*.
For linear independence, suppose $\sum_{i=1}^{n} a_i e_i^* = 0$. Evaluating this zero form on e_j:

$$0 = \left(\sum_{i=1}^{n} a_i e_i^* \right) (e_j) = \sum_{i=1}^{n} a_i \delta_{ij} = a_j.$$

Hence $a_1 = \cdots = a_n = 0$, so $e_1^*, \ldots, e_n^*$ are linearly independent and form a basis of E^*. Consequently, $\dim E^* = n = \dim E$. □

6.2 The Transpose of a Linear Map

Lemma 83: Transpose Linear Map

Let $f : E \to F$ be a linear map between vector spaces. The map defined by:

$$f^* : F^* \longrightarrow E^*, \quad \omega \longmapsto f^*(\omega) := \omega \circ f$$

(where $f^*(\omega)(u) = \omega(f(u))$ for all $u \in E$) is a linear map.

PROOF. Let $\omega_1, \omega_2 \in F^*$ and $\lambda, \mu \in \mathbb{K}$. For any $u \in E$:

$$\begin{aligned} f^*(\lambda\omega_1 + \mu\omega_2)(u) &= ((\lambda\omega_1 + \mu\omega_2) \circ f)(u) = (\lambda\omega_1 + \mu\omega_2)(f(u)) \\ &= \lambda\omega_1(f(u)) + \mu\omega_2(f(u)) = \lambda f^*(\omega_1)(u) + \mu f^*(\omega_2)(u) \\ &= (\lambda f^*(\omega_1) + \mu f^*(\omega_2))(u). \end{aligned}$$

Hence $f^*(\lambda\omega_1 + \mu\omega_2) = \lambda f^*(\omega_1) + \mu f^*(\omega_2)$, so f^* is linear. □

Definition 84: Transpose Map

Let $f : E \to F$ be a linear map. The map $f^* : F^* \to E^*$ defined by $f^*(\omega) = \omega \circ f$ is called the *transpose* (or dual map) of f.

Proposition 85: Matrix Representation of Transpose Map

Let $f : E \to F$ be a linear map of finite dimensional vector spaces. Let $\mathcal{B} = (e_1, \ldots, e_n)$ be a basis of E and $\mathcal{C} = (u_1, \ldots, u_m)$ be a basis of F. Let $A = (a_{ij}) \in \mathrm{Mat}_{m \times n}(\mathbb{K})$ be the matrix of f relative to these bases. Then the matrix of f^* relative to the dual bases

$\mathcal{C}^* = (u_1^*, \ldots, u_m^*)$ and $\mathcal{B}^* = (e_1^*, \ldots, e_n^*)$ is the transpose of A, namely:

$$\mathrm{Mat}_{\mathcal{C}^*,\mathcal{B}^*}(f^*) = {}^tA = (a_{ji})_{n\times m}.$$

PROOF. To find the matrix of f^*, we expand $f^*(u_j^*)$ in terms of the dual basis $(e_1^*, \ldots, e_n^*)$ for $j = 1, \ldots, m$:

$$f^*(u_j^*) = \sum_{k=1}^{n} b_{kj} e_k^*.$$

Thus $B = (b_{ij})_{n\times m}$ is the matrix representing f^*. Evaluating both sides of this equation on the basis vector $e_i \in E$:

$$\mathrm{RHS} = \left(\sum_{k=1}^{n} b_{kj} e_k^*\right)(e_i) = \sum_{k=1}^{n} b_{kj}\delta_{ki} = b_{ij}.$$

For the left-hand side, since $f(e_i) = \sum_{l=1}^{m} a_{li} u_l$:

$$\mathrm{LHS} = f^*(u_j^*)(e_i) = u_j^*(f(e_i)) = u_j^*\left(\sum_{l=1}^{m} a_{li} u_l\right) = \sum_{l=1}^{m} a_{li} u_j^*(u_l) = \sum_{l=1}^{m} a_{li}\delta_{jl} = a_{ji}.$$

Equating both sides gives $b_{ij} = a_{ji}$ for all i, j. Thus $B = {}^tA$. □